Tawfiq Abu Rub

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SUMMARY

Detail-oriented and innovative Mechanical Engineering student, predicted to graduate with First Class Honours, with experience across a range of engineering projects, competitions, and events. Strong interest in robotics, demonstrated through personal, out-of-university projects that have provided valuable hands-on experience in design, development, and problem-solving. Excels under pressure, proactively expanding knowledge and refining skills to effectively overcome complex challenges. A strong team player who emphasises clear and precise communication to promote effective collaboration.

EDUCATION

University of Manchester
2023 - Present

- Mechanical Engineering BEng
- Predicted Average of 74%

University of Manchester
2026 - 2027 (Accepted)

- MSc Robotics

SKILLS

- Solidworks
- FEA
- Engineering Drawings
- DFMA
- FMEA
- 3D Printing
- Python
- Pybullet
- OpenCV
- Matlab
- C++
- Arduino
- Star CCM+
- Abaqus
- CNC Simulator
- MS Suite
- Soldering
- Gant Chart
- Project Management

LANGUAGES

- English (Fluent)
- Arabic (Fluent)

EXPERIENCE

Robotic Arm Drawing System — In Progress Aug 2025 - Present

- Designed (SolidWorks) and **3D-printed** a custom **4DOF robotic arm** driven by an **ESP32-S3** controlling NEMA stepper motors with **custom-designed 25:1 planetary gearboxes.**
- Developed a Python program for **real-time robotic control**, featuring custom **Inverse Kinematics** (IK), point **trajectory interpolation**, and automated wrist-leveling algorithms.
- Utilised **OpenCV** to convert raster graphics into vector paths, and implemented **motion planning** with pen-up/down logic to simulate the 4DOF **URDF** model in **PyBullet** prior to physical execution.
- Prototyping a **teleoperated "Leader" arm** and a **dual-camera vision** system to record rich **visual state-action pairs**, establishing the groundwork to train an **Imitation Learning** model in **PyTorch**.

Mechanical Engineering Intern - Gulf Glass Manufacturing Company (GGMC), Kuwait Jul 2025 - Aug 2025

- Supported the maintenance department in **preventative maintenance** and troubleshooting of **glass production machinery**, including batch plant, furnace and IS Machines.
- Worked with a large **multidisciplinary team of engineers** and technicians in a fast-paced industrial environment.

2024 IMechE Design Challenge Sept 2023 - Dec 2023

- Designed and assembled a fully mechanical autonomous vehicle capable of tracking and repeating traversed movements with **98% positional accuracy**.
- Applied project management techniques like **Network Diagrams** and **Gantt Charts** to ensure project delivery within time and budget constraints.

Formula Student Engineer Oct 2024 - Present

- Engineered a new brake disc with enhanced **airflow**, increased **heat dissipation**, and a **15% weight reduction**, ensuring full compliance with the 2024 Formula Student regulations.
- Conducted **FMEA analysis** validating the brake disc's ability to endure sustained high-temperature friction and the brake caliper's **maximum loading conditions**.

Drilling Machine Design Sept 2024 - Dec 2024

- Led a dynamic team of engineers, managing task distribution and coordination amongst team members to optimise workflow and project delivery, resulting in a final project score of 82%.
- Performed **deflection analysis**, minimising the total deflection at the drill bit to **100µm.**
- Designed and optimised a belt and pulley **drivetrain assembly**, ensuring proper **belt tensioning** to minimise power loss.